

Context

Companies use servers of all types, and an Information Technology Professional is expected to be able to setup, optimize, and maintain servers of various operating systems and functions.

For this project, you have been tasked with setting up a web server to host a small website. A single server has been allocated for this use, which you will simulate using Oracle VirtualBox. You may choose to use any version of Windows server, or any distribution of Linux as your operating system. All other required software will be chosen using open source or trial versions for setting up the actual server instance. The web developer will provide you with the website files in a .zip file to load on the server in the Final Project Draft assignment.

This assignment will be done in phases.

- Week 9: Final Project Part 1 - Proposal
 - You will submit your proposal with your choice of operating system and a list of open-source or trial software that you will be installing to make the web server operational. This will include the justification for your choices.
- Week 11: Final Project Part 2 - Draft
 - You will submit screenshots of the operational web server administrative screens and output, based on feedback received on the proposal.
- Week 14: Final Project Part 3 - Submission
 - You will submit the updated documents from weeks 9 and 11 and a three- to five-minute video describing the installation, optimization, and current operation of the web server that was simulated with Oracle VirtualBox.

Steps

For this assignment: Final Project Part 3, you will need to utilize the feedback you received from your instructor for Weeks 9 and 11 assignments.

1. Utilizing the feedback from your proposal and draft assignments, update your operating system and software list as required. Save that document in DOCX or PDF format.
2. Utilizing the feedback from your draft assignment, update or modify your web server instance as required. Retake the screenshots required for the draft assignment and save the compiled document in DOCX or PDF format.

My Ubuntu Server

Software	Version	Type	Purpose
Apache HTTP Server	2.4x	Open Source	Handles HTTP requests and presents pages to viewers
MySQL Server	8.0	Open Source	Database management to store website data
PHP	8.1	Open Source	Scripting language for making web pages (server-side)
phpMyAdmin	5.2	Open Source	Interface for managing MySQL Databases
OpenSSH	10.2	Open Source	Allows remote access and management of the server
Vim	9.2	Open Source	Command line text editor. (Nano also could work but I prefer Vim)
Git	2.53.0	Open Source	Version control system for managing and pushing website code
Fail2Ban	1.1.0	Open Source	Log management while also banning suspicious IPs
Cerbot	5.4.0	Open Source	Allows free SSL/TLS certificates for HTTPS encryption
Nginx	1.29.2	Open Source	Better performance, load handling, and static file delivery in front of Apache.
UFW firewall	26.04	Open source	Simple firewall management and

			making controlling ports much easier
Redis	8.4.2	Open Source	Acts as a database or application cache

Apache server status

```
davidfinal@finalproject:~$ systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-04-09 14:05:36 UTC; 8min ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 15709 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
  Main PID: 15712 (apache2)
    Tasks: 7 (limit: 2263)
   Memory: 11.8M (peak: 12.0M)
      CPU: 262ms
   CGroup: /system.slice/apache2.service
           └─15712 /usr/sbin/apache2 -k start
             └─15715 /usr/sbin/apache2 -k start
               └─15716 /usr/sbin/apache2 -k start
                 └─15717 /usr/sbin/apache2 -k start
                   └─15718 /usr/sbin/apache2 -k start
                     └─15719 /usr/sbin/apache2 -k start
                       └─15885 /usr/sbin/apache2 -k start

Apr 09 14:05:36 finalproject systemd[1]: Starting apache2.service - The Apache HTTP Server...
Apr 09 14:05:36 finalproject apachectl[15711]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set t
Apr 09 14:05:36 finalproject systemd[1]: Started apache2.service - The Apache HTTP Server.
```

SSH

```
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
canning processes...
canning candidates...
canning linux images...

unning kernel seems to be up-to-date.

estarting services...

ervice restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

o containers need to be restarted.

ser sessions running outdated binaries:
davidfinal @ session #1: login[729]
davidfinal @ user manager service: systemd[883]

o VM guests are running outdated hypervisor (qemu) binaries on this host.
avidfinal@finalproject:~$ systemctl status ssh
ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: inactive (dead)
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
```

(Git, Vim, Fail2bam and Certbot) installation

```
Ubuntu Web Server [Running] - Oracle VirtualBox
File Machine View Input Devices Help

"1990349000 test failed,dns.ch", "\xstest <F-ID>\S+</F-ID>"
/usr/lib/python3/dist-packages/fail2ban/tests/fail2banregextestcase.py:435: SyntaxWarning: invalid escape sequence '\S'
'^'+prefix+<F-ID>User <F-USER>\S+</F-USER></F-ID> not allowed\n'
/usr/lib/python3/dist-packages/fail2ban/tests/fail2banregextestcase.py:443: SyntaxWarning: invalid escape sequence '\S'
'^'+prefix+ 'User <F-USER>\S+</F-USER> not allowed\n'
/usr/lib/python3/dist-packages/fail2ban/tests/fail2banregextestcase.py:444: SyntaxWarning: invalid escape sequence '\d'
'^'+prefix+ Received disconnect from <F-ID><ADDR> port \d+</F-ID>'
/usr/lib/python3/dist-packages/fail2ban/tests/fail2banregextestcase.py:451: SyntaxWarning: invalid escape sequence '\s'
test_variants('common', prefix="\s\S+ sshd(<F-MLFID>\d+</F-MLFID>\)\S+"))
/usr/lib/python3/dist-packages/fail2ban/tests/fail2banregextestcase.py:537: SyntaxWarning: invalid escape sequence '\['
'common[pre]regex="\s\S+(<F-MLFID>\d+</F-MLFID>\)\s\S+ connect <F-CONTENT>.+</F-CONTENT>)"$'
/usr/lib/python3/dist-packages/fail2ban/tests/servertestcase.py:1375: SyntaxWarning: invalid escape sequence '\s'
" { nft -a list chain inet f2b-table f2b-chain | grep -oP '@addr-set-j-w-nft-mp\S+.*\S+Khandle\S+(\d+)\$'; } | while read -r hdl; do '",
/usr/lib/python3/dist-packages/fail2ban/tests/servertestcase.py:1378: SyntaxWarning: invalid escape sequence '\s'
" { nft -a list chain inet f2b-table f2b-chain | grep -oP '@addr6-set-j-w-nft-mp\S+.*\S+Khandle\S+(\d+)\$'; } | while read -r hdl; do '",
/usr/lib/python3/dist-packages/fail2ban/tests/servertestcase.py:1421: SyntaxWarning: invalid escape sequence '\s'
" { nft -a list chain inet f2b-table f2b-chain | grep -oP '@addr-set-j-w-nft-ap\S+.*\S+Khandle\S+(\d+)\$'; } | while read -r hdl; do '",
/usr/lib/python3/dist-packages/fail2ban/tests/servertestcase.py:1424: SyntaxWarning: invalid escape sequence '\s'
" { nft -a list chain inet f2b-table f2b-chain | grep -oP '@addr6-set-j-w-nft-ap\S+.*\S+Khandle\S+(\d+)\$'; } | while read -r hdl; do '",
Created symlink /etc/systemd/system/multi-user.target.wants/fail2ban.service → /usr/lib/systemd/system/fail2ban.service.
Setting up python3-pynotify (0.9.6-2ubuntu1) ...
Setting up python3-acme (2.9.0-1) ...
Setting up python3-certbot (2.9.0-1) ...
Setting up certbot (2.9.0-1) ...
Created symlink /etc/systemd/system/timers.target.wants/certbot.timer → /usr/lib/systemd/system/certbot.timer.
Setting up python3-certbot-apache (2.9.0-1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
/etc/needrestart/nrstart.d/abus.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
davidfinal @ session #1: login(729)
davidfinal @ user manager service: systemd[883]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
davidfinal@finalproject:~$ sudo apt install git vim fail2ban certbot python3-certbot-apache -y
```

```
Ubuntu Web Server [Running] - Oracle VirtualBox
File Machine View Input Devices Help

davidfinal@finalproject:/home$ cd ..
davidfinal@finalproject:/home$ ls -l /home/davidfinal
-rw-r--r-- 1 davidfinal 4096 Nov 19 12:00 website.zip
davidfinal@finalproject:/home$ cd ~
davidfinal@finalproject:~$ cd /home/davidfinal
davidfinal@finalproject:~$ dir
website.zip
davidfinal@finalproject:~$ sudo apt install unzip -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  zip
The following NEW packages will be installed:
  unzip
0 upgraded, 1 newly installed, 0 to remove and 4 not upgraded.
Need to get 174 kB of archives.
After this operation, 384 kB of additional disk space will be used.
Get:1 http://us.archive.ubuntu.com/ubuntu noble-updates/main amd64 unzip amd64 6.0-28ubuntu4.1 [174 kB]
Fetched 174 kB in 2s (90.8 kB/s)
Selecting previously unselected package unzip.
Reading database ... 99130 files and directories currently installed.)
Preparing to unpack .../unzip_6.0-28ubuntu4.1_amd64.deb ...
Unpacking unzip (6.0-28ubuntu4.1) ...
Setting up unzip (6.0-28ubuntu4.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
davidfinal@finalproject:~$ unzip website.zip
Archive:  website.zip
  creating: website/images/
  inflating: website/images/background.png
  inflating: website/index.html
  inflating: website/styles.css
davidfinal@finalproject:~$ sudo mv website/* /var/www/html/
davidfinal@finalproject:~$ sudo systemctl restart apache2
davidfinal@finalproject:~$ cd /var/www/html/
davidfinal@finalproject:/var/www/html$ dir
images index.html styles.css
davidfinal@finalproject:/var/www/html$
```

Nginx install and status:

```
Ubuntu Web Server [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
0 packages can be upgraded. Run 'apt list --upgradable' to see them.
avidfinal@finalproject:~$ sudo apt install nginx
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 9 not upgraded.
134690,787215] watchdog: BUG: soft lockup - CPU#1 stuck for 26s! [snap:6055]

Need to get 565 kB of archives.
After this operation, 1,596 kB of additional disk space will be used.
Get:1 http://us.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.6 [43.5 kB]
Get:2 http://us.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.6 [521 kB]
Fetched 565 kB in 18s (31.8 kB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
Reading database ... 100470 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.24.0-2ubuntu7.6_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.6) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.24.0-2ubuntu7.6_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.6) ...
Setting up nginx-common (1.24.0-2ubuntu7.6) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
Could not execute systemctl: at /usr/bin/deb-systemd-invoke line 148.
Setting up nginx (1.24.0-2ubuntu7.6) ...
Not attempting to start NGINX, port 80 is already in use.
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

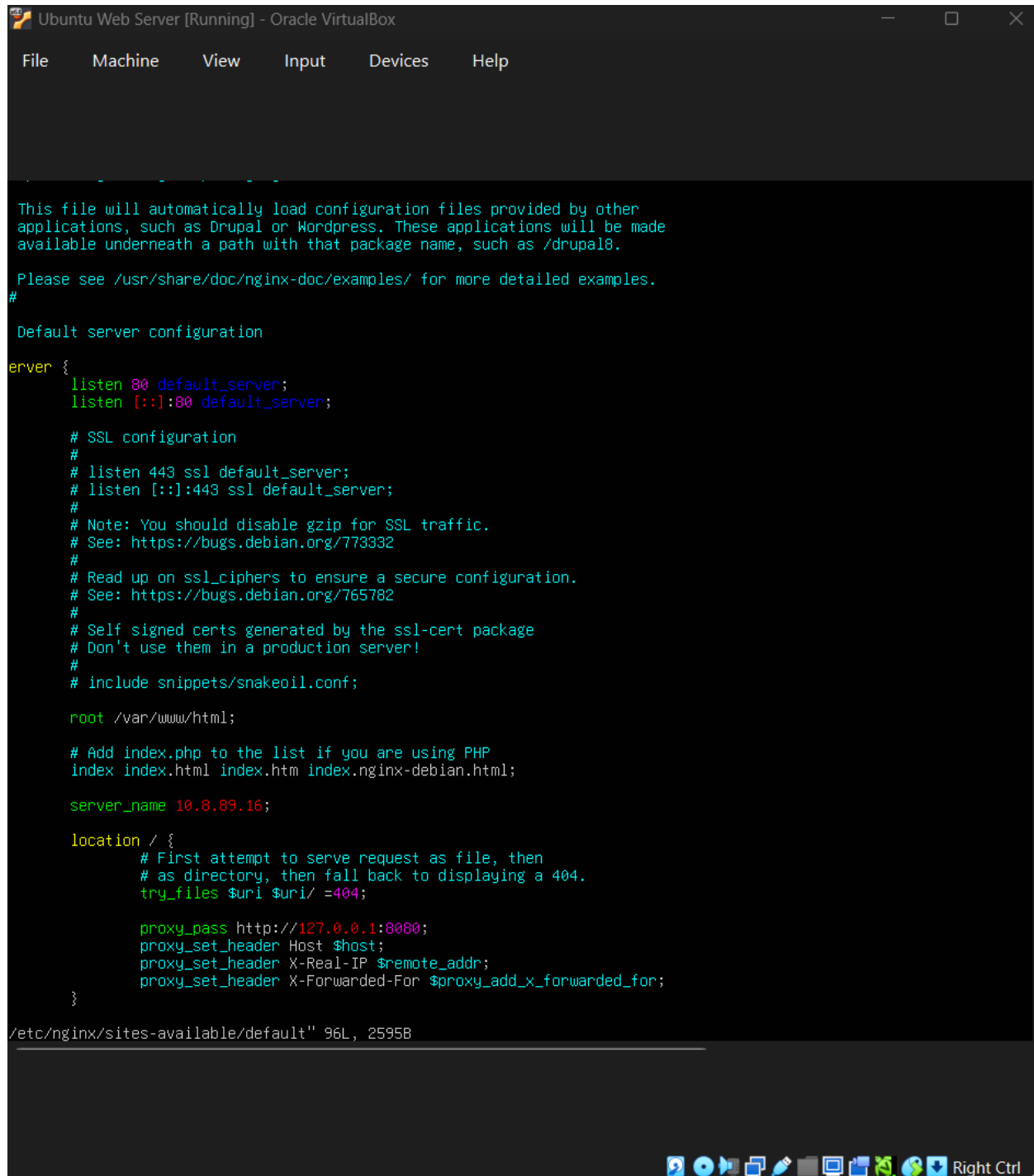
No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
avidfinal@finalproject:~$

avidfinal@finalproject:~$ sudo systemctl status nginx
nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-04-13 22:02:27 UTC; 1min 18s ago
     Docs: man:nginx(8)
  Process: 2236 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
  Process: 2237 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
 Main PID: 2239 (nginx)
    Tasks: 3 (limit: 2262)
   Memory: 2.3M (peak: 2.5M)
      CPU: 19ms
   CGroup: /system.slice/nginx.service
           └─2239 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─2240 "nginx: worker process"
               └─2241 "nginx: worker process"

apr 13 22:02:27 finalproject systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy se
apr 13 22:02:27 finalproject systemd[1]: Started nginx.service - A high performance web server and a reverse proxy ser
avidfinal@finalproject:~$ _
```

Setup for Nginx



```
Ubuntu Web Server [Running] - Oracle VirtualBox
File Machine View Input Devices Help

This file will automatically load configuration files provided by other
applications, such as Drupal or Wordpress. These applications will be made
available underneath a path with that package name, such as /drupal8.

Please see /usr/share/doc/nginx-doc/examples/ for more detailed examples.
#

Default server configuration
server {
    listen 80 default_server;
    listen [::]:80 default_server;

    # SSL configuration
    #
    # listen 443 ssl default_server;
    # listen [::]:443 ssl default_server;
    #
    # Note: You should disable gzip for SSL traffic.
    # See: https://bugs.debian.org/773332
    #
    # Read up on ssl_ciphers to ensure a secure configuration.
    # See: https://bugs.debian.org/765782
    #
    # Self signed certs generated by the ssl-cert package
    # Don't use them in a production server!
    #
    # include snippets/snakeoil.conf;

    root /var/www/html;

    # Add index.php to the list if you are using PHP
    index index.html index.htm index.nginx-debian.html;

    server_name 10.8.89.16;

    location / {
        # First attempt to serve request as file, then
        # as directory, then fall back to displaying a 404.
        try_files $uri $uri/ =404;

        proxy_pass http://127.0.0.1:8080;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    }
}

/etc/nginx/sites-available/default" 96L, 2595B

Right Ctrl
```

Changed Apache2 port (so that it works with nginx):

```
Ubuntu Web Server [Running] - Oracle VirtualBox
File Machine View Input Devices Help

# have to change the VirtualHost statement in
# /etc/apache2/sites-enabled/000-default.conf

Listen 8080

<IfModule ssl_module>
    Listen 443
</IfModule>

<IfModule mod_gnutls.c>
    Listen 443
</IfModule>
```

Apache2 status:

```
davidfinal@finalproject:~$ sudo systemctl status apache2
• apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-04-13 21:58:06 UTC; 7min ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 2207 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
  Main PID: 2210 (apache2)
    Tasks: 6 (limit: 2262)
   Memory: 15.9M (peak: 16.2M)
      CPU: 179ms
   CGroup: /system.slice/apache2.service
           └─2210 /usr/sbin/apache2 -k start
             └─2212 /usr/sbin/apache2 -k start
               └─2213 /usr/sbin/apache2 -k start
                 └─2214 /usr/sbin/apache2 -k start
                   └─2215 /usr/sbin/apache2 -k start
                     └─2216 /usr/sbin/apache2 -k start

Apr 13 21:58:06 finalproject systemd[1]: Starting apache2.service - The Apache HTTP Server...
Apr 13 21:58:06 finalproject apachectl[2209]: AH00558: apache2: Could not reliably determine the server's fully qualif
Apr 13 21:58:06 finalproject systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-20/20 (END)
```

Ufw install and setup:

```
davidfinal@finalproject:~$ sudo apt install ufw
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ufw is already the newest version (0.36.2-6).
0 upgraded, 0 newly installed, 0 to remove and 9 not upgraded.
davidfinal@finalproject:~$ sudo ufw allow 'Apache Full'
Skipping adding existing rule
Skipping adding existing rule (v6)
davidfinal@finalproject:~$ sudo ufw enable
Firewall is active and enabled on system startup
davidfinal@finalproject:~$ sudo ufw status
Status: active

To Action From
--
OpenSSH ALLOW Anywhere
Apache Full ALLOW Anywhere
OpenSSH (v6) ALLOW Anywhere (v6)
Apache Full (v6) ALLOW Anywhere (v6)

davidfinal@finalproject:~$ _
```

